Zilla-64: A Bangla Handwritten Word Dataset Of 64 Districts` Name of Bangladesh and Recognition Using Holistic Approach

Md Ali Azad
Computer Science and Engineering
Sylhet Engineering College
Sylhet-3100, Bangladesh
aliazad969@gmail.com

Hijam Sushil Singha
Computer Science and Engineering
Sylhet Engineering College
Sylhet-3100, Bangladesh
hijamsingha@gmail.com

Md Mahadi Hasan Nahid
Computer Science and Engineering
Shahjalal University of Science and Technology
Sylhet-3114, Bangladesh
nahid-cse@sust.edu

***Abstract*— Bangla Handwritten Word Recognition (BHWR) is a very challenging task due to the high curvature nature of the character, overlapping between characters, and flourishes in the writing style of Bangla Handwritten Word (BHW). Despite the importance and challenges of BHWR, the handwritten word dataset of Bangla is very few. In this paper, a new challenging Bangla word dataset which, we called 'Zilla-64', is introduced, and also the preparation of the dataset is discussed. To the best of our knowledge, this is the first well-labeled Bangla word dataset. This dataset can be used for gender, age, and education level handwritten related researches. Deep learning shows tremendous success in the handwritten recognition area. Because of the popularity of deep learning methods and for testing the performance of the dataset, a holistic approach based, Deep Convolutional Neural Network (DCNN) is applied on the dataset and achieved 93.30% accuracy. The Zilla-64 dataset is made publicly available at this link https://github.com/MahadiHasanNahid/Zilla-64-Dataset.**

Keywords*— *Convolutional Neural Network, Holistic Approach, Bangla Handwritten Word, Bangla Handwritten Word Recognition, Zilla-64.

I. INTRODUCTION

Bangla is the national and official language of Bangladesh, and it is also the official language of some of India's states, such as West Bengal and Tripura [1]. Bangla handwritten recognition, especially for character and word recognition, has drawn considerable attention in these decades. Many types of research have been performed to recognize Bangla Handwritten Character (BHC), but for the Bangla Handwritten Word Recognition (BHWR), the number of studies is less than Bangla Handwritten Character Recognition (BHCR). Handwritten Word Recognition can be done using analytical and holistic approaches. The analytical word recognition needs proper segmentation of character in a word, which is very challenging in the recognition area, especially in BHWR. In contrast, holistic word recognition is a segmentation-free approach where a word is considered a single and inseparable unit, and it attempts to recognize words as a whole using their representational features without any character segmentation [1]. It should be mentioned that the human mind acknowledges a word in a holistic manner, which is reported by this paper [2]. In most cases, researchers seek a dataset to apply and verify the performance of their model. An appropriate dataset is also necessary for proper research. English, Arabic, and other languages have significant resources for handwritten word datasets, but the Bangla handwritten word dataset has minimal resources [3].

Recent works in BHCR show outstanding performance using deep learning methods. Deep learning methods like Convolutional Neural Network (CNN) have proven their superiority in the BHCR area [4]-[11]. So, CNN-based experiments should be conducted in the BHWR area. The holistic approach using CNN could be a good model than others because it performs the operations holistically. The drawback of this approach is that it is only useful with a fixed or limited lexicon. Inspire by the success of CNN in BHCR, we used a CNN model in this paper to test the performance of our new dataset. Major problems like 'segmentation ambiguity,' i.e., deciding where to segment the word image, and 'variability of segment shape,' i.e., determining the identity of each segment of an analytical approach, can be vanished using a holistic approach [12]. One of the main applications of handwritten word recognition using a holistic approach is to validate a limited feature handwritten form. The proposed work can be used for the recognition of the district field in the address bar of a handwritten paper form that has been written in Bangla. Also, the introduced dataset can be challenging work for Bangla's handwritten character segmentation analysis. In the handwritten recognition area, the Bangla language has more curvature nature and overlapping characters, mostly compound characters, which makes it challenging than the other languages. Some challenging words show in Figure 1, which is very hard to recognize using an analytical approach.

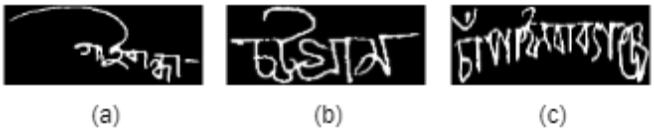


Fig. 1: (a) flourishes in writing style, (b) overlapping among characters, (c) curvature nature of a word.

First and Foremost, for the dull research progress on BHWR is the unavailability of proper databases. The primary purpose of this paper is to introduce a new well-prepared BHW dataset with details preparation of the dataset that will

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facilitate and support researchers for future studies. Each sample of the dataset has been labeled by its writer's gender, age, and education level. This dataset will be a significant contribution to the BHW dataset area.

II. Existing Datasets And Systems

The English literature enriches with a vast collection of handwritten image databases. Mentionables, The MNIST [13] and CENPARMI [14] databases contain handwritten numerals, the I-AM [15] database is a big collection of handwritten sentences, and the CEDAR [16] database has handwritten all types of handwritten text needed for postal address recognition. The CENPARMI database also contains handwritten characters and word samples of Urdu, Farsi, and Arabic languages. The publicly available datasets for the handwritten recognition in Bangla literature are ISI [17], CMATERdb [3], Banglalekha-Isolated [18], Ekush [19], AIBangla [20], and BanglaWriting [21] datasets. Among them, the CMATERdb database only contains Bangla handwritten word images, which are not well-labeled and well-prepared. This database holds two BHW datasets. The BanglaWriting dataset contains only single-page handwriting random documents of 260 individuals. The dataset is not well-organized and well-prepared for handwritten word recognition. The CMATERdb 2.1.1 has 5000 words sample for character segmentation purposes, and CMATERdb 2.1.2 has 18000 words sample of 120 classes for the handwritten holistic recognition process. The CMATERdb 2.1.1 and CMATERdb 2.1.2 datasets include over-cut, improper alignment, and noisy samples. A model efficiency decreases with the increase of these properties sample in a dataset. Mostly, noisy and over-cut samples are challenging to determine in any recognition system, even in human eyes. Figure 2 shows some error samples from the two datasets of CMATERdb.

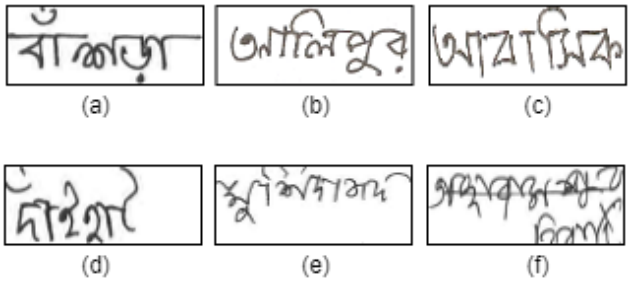


Fig. 2: Images of CMATERdb 2 dataset. (a, b, c) sample images, (d) word doesn't align in the center, (e) Over-cut of a word, (f) Noisy image.

The two datasets (Ekush with 336,351 samples for 112 classes and AIBangla with 330,314 samples for 221 classes) of BHC have drawn significant attention with a considerable massive number of samples in the Bangla literature. Also, the other three datasets (ISI, CMATERdb 3.1, BanglaLekha-Isolated) have immense contributions. The Ekush dataset is considered the largest dataset in the Bangla handwritten recognition area. So, it can be said that BHC datasets' resources are enormous compared to the BHW dataset. This is one of the main reasons that the researches are higher for the BHCR than the BHWR. The lack of a proper dataset of handwritten word images is a bottleneck for the research in the field of Bangla handwritten word recognition.

In recent years, many efforts have been undertaken using the holistic approach to recognizing BHW in various scripts. This approach is studied for BHWR in these papers [1], [12], [22]-[27]. Histogram of Oriented Gradients (HOG) as the feature set with Multi-Layer perceptron (MLP) as the classifier is used to recognize BHW in this paper [1]. They [22] applied five well-known supervised learning-based classifiers (Naive Bayes, MLP, Bagging, Dagging, and Support Vector Machine (SVM)) and achieved final accuracy using MLP classifier with elliptical features. They [12] tried to recognizing BHW using seven classifiers (MLP, SVM, Naïve Bayes, Dagging, K-nearest neighbor(KNN), Logit-boost, and Bagging) with convex hull features and concentric rectangles feature extraction. All of these experiments [12] showed comparatively good performances in three different datasets. Hidden Markov Model (HMM) [23] based on sub-stroke level feature is proposed for the online BHWR. The work described in this paper [24] introduced the Tetragonal feature along with the Vertical pixel density histogram-based feature in Bangla literature. They used these two features as a shape-based feature descriptor, and finally, the recognition process is done by the MLP and SVM classifier. Recently, a few research types based on the CNN model, such as H-WordNet [25], lottery ticket hypothesis-based pruned CNN model [26], and CNN-based transfer learning [27], is used for BHWR. H-WordNet shows outstanding performance for the recognition of 120 classes. All of these mentioned works tried various techniques for the handwritten recognition process. Most of them used a supervised-based machine learning technique. Unsupervised-based practices didn't yet apply in the BHWR area as a holistic approach. Above all, mentioned works' performance is mostly satisfactory. But, Most of them used the same database CMATERdb and performed their operations in two different datasets. Variation of the database is needed to prove the stability and robustness of a model.

III. The Zilla-64 Dataset

The Zilla-64 dataset is named according to the 64 districts' names of Bangladesh. This dataset is written by 34 writers of different gender, age, and education level.

A. Data collection

Writers of the Zilla-64 dataset were selected from various ages, gender, and educational background. The proposed dataset of Bangla handwritten words was collected from 34 persons, where 58.8% were male and 41.2% were female. We made two types of forms to collect the data. The single word form is for the writer who wants to write each word for once, and the multiple words form is for the writer who wants to write each word five times. The single word form has two pages with a total of 64 classes (36 classes on the first page and the rest on the second page). On the other hand, the multiple word form has five pages of 5 empty columns with 64 classes (12 classes on the first page and the rest of the four pages have equal 13 classes). Cell with the border was created in the A4-form to separate and crop the word sample. Participants were asked to write given printed words carefully and independently. After filling the forms, the collector verifies each page to ensure that it has been correctly written with given words.



Fig. 3: Single handwriting form (first page)

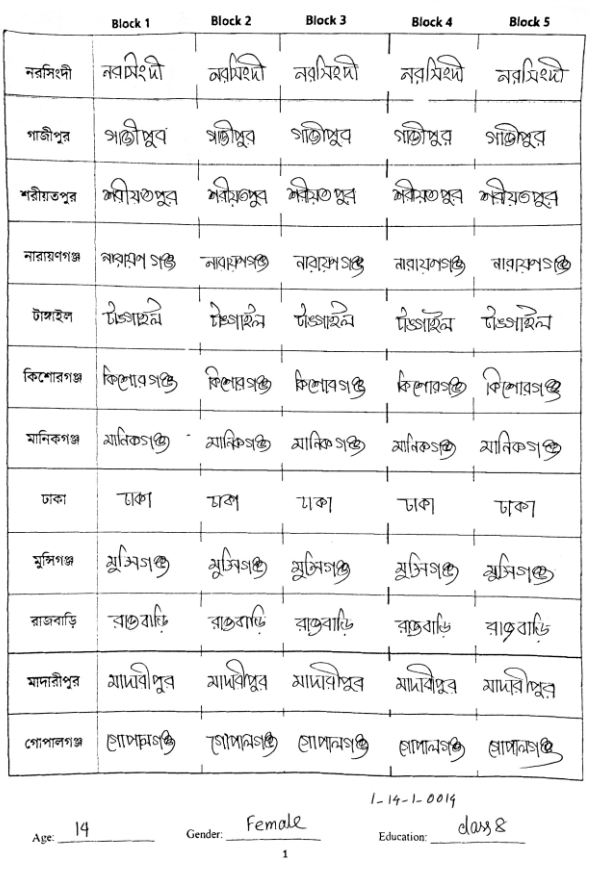


Fig. 4: Multiple handwriting form (first page)

TABLE I: WORD CLASSES OF 'ZILLA-64'

1	নরসিংদী	26	কক্সবাজার	51	ভোলা
2	গাজীপুর	27	খাগড়াছড়ি	52	বরগুনা
3	শরীয়তপুর	28	বান্দরবান	53	সিলেট
4	নারায়ণগঞ্জ	29	সিরাজগঞ্জ	54	মৌলভীবাজার
5	টাঙ্গাইল	30	পাবনা	55	হবিগঞ্জ
6	কিশোরগঞ্জ	31	বগুড়া	56	সুনামগঞ্জ
7	মানিকগঞ্জ	32	রাজশাহী	57	পঞ্চগড়
8	ঢাকা	33	নাটোর	58	দিনাজপুর
9	মুন্সিগঞ্জ	34	জয়পুরহাট	59	লালমনিরহাট
10	রাজবাড়ি	35	চাঁপাইনবাবগঞ্জ	60	নীলফামারী
11	মাদারীপুর	36	নওগাঁ	61	গাইবান্ধা
12	গোপালগঞ্জ	37	যশোর	62	ঠাকুরগাঁও
13	ফরিদপুর	38	সাতক্ষীরা	63	রংপুর
14	শেরপুর	39	মেহেরপুর	64	কুড়িগ্রাম
15	ময়মনসিংহ	40	নড়াইল		
16	জামালপুর	41	চুয়াডাঙ্গা		
17	নেত্রকোনা	42	কুষ্টিয়া		
18	কুমিল্লা	43	মাগুরা		
19	ফেনী	44	খুলনা		
20	ব্রাহ্মণবাড়িয়া	45	বাগেরহাট		
21	রাঙ্গামাটি	46	ঝিনাইদহ		
22	নোয়াখালী	47	ঝালকাঠি		
23	চাঁদপুর	48	পটুয়াখালী		
24	লক্ষ্মীপুর	49	পিরোজপুর		
25	চট্টগ্রাম	50	বরিশাল		

B. Preparation of the dataset

Bangla handwritten district names of Bangladesh 64 Zilla were extracted using various steps. All steps are shown and described via a block diagram in Figure 5. The collections were scanned in jpg format through an android app using a smartphone camera. After the scanning process, GIMP 2.10.20 [28] software was used to cropping and removing lines, draw slicing lines, and segment the words from the scanned image. We used python programming language for class labeling, binarization, noise reduction, and fit to size process. Black background with white foreground was maintained to produce all samples of our dataset. For this, the Bitwise-Not operation was performed on the grayscale images.

The Otsu process was applied to perform automatic image thresholding. We used the Binary-Otsu to create bitonal (binary) images based on the threshold values. The Non-local Means denoising technique was applied for the noise reduction of images. This filtering process takes a mean of all pixels from an image. Then, it is weighted by how related those pixels are to the target pixel. Fit to size is one kind of method of geometric transformation. It was used to correct the misaligned handwritten images. For this, first, we identify the co-ordinations of handwritten foreground white regions in the grayscale image. Then, these coordinates were used to crop the original grayscale image. In the final step, black padding was added to preserve the aspect ratio for all samples.

All collected forms were labeled before scanning. The extracted sample from each form was saved as a png format,

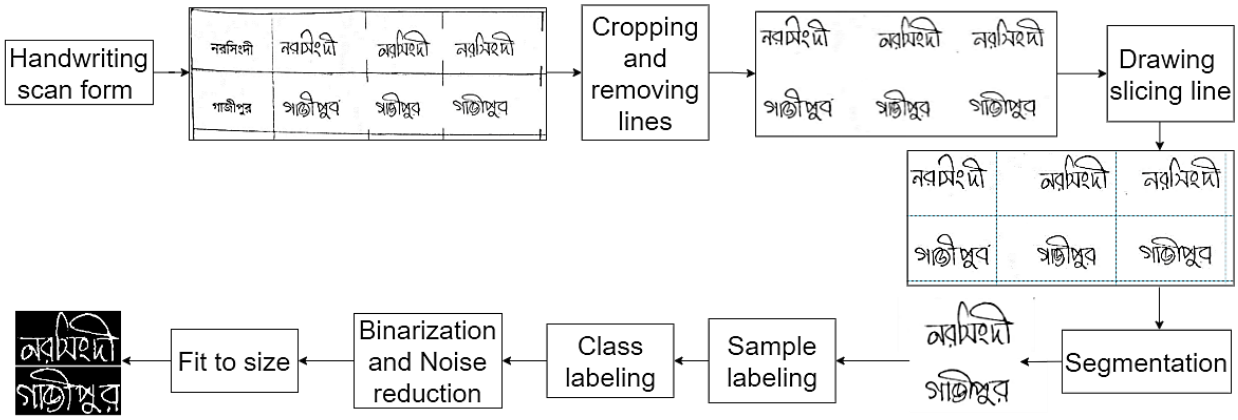


Fig. 5: Block diagram for the preparation of Zilla-64 dataset

and each sample was given its form unique id or name that represents the writer's information of that sample. We collected the age, gender, and educational background information of writers for research and data labeling purposes. The unique id or name of an image in Zilla-64 is made using the following criteria, which are described in Table II. An underscore separates the information of our proposed dataset sample.

TABLE II: DATA LABELING CRITERIA

Gender(id)	Age	Education level(id)	Form-id	Word position
Male(0)	Age of a person	Primary(0)	A unique id of the form	The position of the word in the form
		High School(1)		
Female(1)		College(2)		
		University(3)		

C. Overview of 'Zilla-64'

The dataset has 4480 handwritten word samples. For the comparison of male-female handwriting, our proposed dataset has also contained male-female divided datasets. Since the dataset is about the 64 districts name of Bangladesh, the dataset has 64 classes, and all classes have equally distributed 70 images of the handwritten word.

TABLE III: A BRIEF SUMMARY OF THE 'ZILLA-64' DATASET

Number of Writers	34 persons
Age range	11-55 y (years)
Age ratio	10y-20y: 52.9%
	21y-30y: 29.4%
	31y-40y: 8.8%
	41y-60y: 8.8%
Education level range	Primary to University
Education level ratio	Primary: 5.9%
	High school: 17.6%
	College: 26.5%
	University: 50.0%
Total class	64
Number of samples per class	70 samples
Total samples	4480 samples

TABLE IV: CHARACTERS LIST WITH NUMBER OF OCCURRENCES AND FREQUENCY IN THE 'ZILLA-64' CLASSES

Character Type	List	Occurrences	Frequency (%)
Modifiers	া, ি, ী, ু, ে, ো, ৌ, র-ফলা	153	35.17%
Vowels	ই, ও	7	1.61%
Consonants	ক, খ, গ, চ, ছ, জ, ঝ, ট, ঠ, ড, ঢ, ণ, ত, দ, ন, প, ফ, ব, ভ, ম, য, র, ল, শ, স, হ, ড়, য়, ং, ঁ	253	58.16%
Compound Characters	ঞ্জ, ঙ্গ, ন্স, ত্র, ল্ল, ষ্ম, ক্ষ্ম, ক্ষ, ট্ট, ক্স, ষ্ট, ন্ধ	22	5.06%

TABLE V: BHW PUBLICLY AVAILABLE DATASETS COMPARISON

Dataset	Classes	Total samples	Samples labeling
CMATERdb 2.1.1	Unclassified	5000	No
CMATERdb 2.1.2	120	18000	No
Zilla-64	**64**	**4480**	**Yes**

Zilla-64 is not only for handwritten recognition, but also it helps to find individual information. The usages of the proposed dataset in the computer vision and handwriting biometrics areas are significant. The dataset is suitable for machine learning and deep learning models using offline approaches. In handwriting biometrics, the dataset can be used for age, gender, or education-level predictions. It also can be used for handwriting segmentation and recognition process in the computer vision area. Mostly, it is useful for testing a deep learning model efficiency.

D. Error correction

Some of the persons made a mistake in writing the word correctly. The class names of Zilla-64 are mostly constructed with compound characters, and most of the terms are longer. So the writers often made mistakes, such as putting a dot below the character, clear writings of the compound characters, etc. There were a few words that were misclassified by the writers. As the misclassified samples were few, we made the writers correct them in the form collection time. We re-examined our dataset more than five times to ensure an error-free dataset. Most of the steps were taken manually in order to provide the quality and correctness of the proposed Zilla-64 dataset.

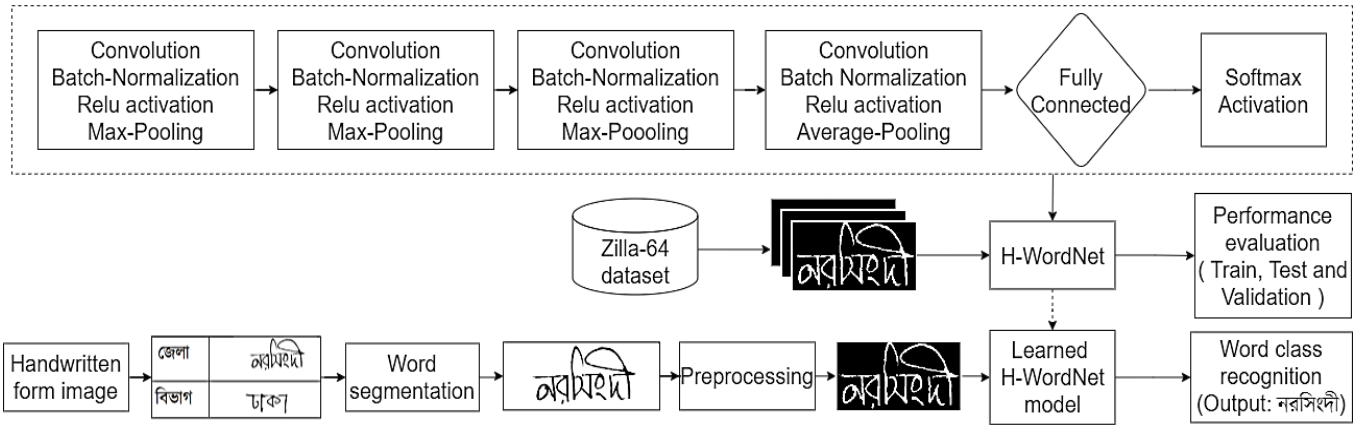


Fig. 6: Block diagram of holistic approach for the recognition of handwritten word

IV. Recognition System

We used the H-WordNet for the recognition of the proposed Zilla-64 dataset. The H-WordNet shows tremendous performance for recognizing handwritten cities names of the CMATERdb 2.1.2 dataset and it is explained in this paper [25]. The model is designed with only four Convolutional layers and one Fully Connected dense layer. The performance of the model is linked by using the three Max-Pooling layers and one Average-Pooling layer. The overall block diagram of the holistic approach using the H-WordNet model shows in Figure 6.

TABLE VI: LAYERS SUMMARY OF H-WORDNET FOR THE RECOGNITION OF 'ZILLA-64' DATASET

Layer (Type)	Output shape	Parameters
Convolution 1	(None, 90, 390, 16)	1952
Batch-Normalization 1	(None, 90, 390, 16)	64
Max-Pooling 1	(None, 45, 195, 16)	0
Convolution 2	(None, 39, 189, 32)	25120
Batch-Normalization 2	(None, 39, 189, 32)	128
Max-Pooling 2	(None, 19, 94, 32)	0
Convolution 3	(None, 15, 90, 64)	51264
Batch-Normalization 3	(None, 15, 90, 64)	256
Max-Pooling 3	(None, 7, 45, 64)	0
Convolution 4	(None, 5, 43, 128)	73856
Batch-Normalization 4	(None, 5, 43, 128)	512
Average-Pooling	(None, 2, 21, 128)	0
Flatten	(None, 5376)	0
Dense	(None, 64)	344128
Activation	(None, 64)	0
Total parameters		**497,280**
Trainable parameters		**496,800**
Non-trainable parameters		**480**

The four Convolutional blocks with Batch-Normalization, Relu-Activation, and Pooling layers serve as a feature learning process. The classification process is done using the Fully-Connected layer and Softmax-Activation. The learned representation from H-WordNet trained by the Zilla-64 dataset can be saved, and it can be helpful for the recognition of handwritten Bangla Zilla words in a handwritten scanned form image. The framework of this model is learned using the back-propagation method.

A. Train, Test, and Validation

All input images were resized to 100 x 400 (width x height), with their aspect ratio retained. Categorical cross-entropy as loss and Adam as an optimizer is used for the compilation of the model. We did not increase the samples per epoch or validation per epoch during the training and testing to improve the model's performance in a few iterations.

B. Performance Evaluation

Implementation of the H-WordNet model was done by using the Keras [29] framework with the TensorFlow [30] backend. For the evaluation process, 80% images for training, 10% images for testing, and the rest of the images for validation purposes was randomly selected from the proposed dataset. A batch size of 20 for 50 epochs was applied for evaluating the performance. After the iterations, the H-WordNet achieved 93.30% accuracy in testing, 96.21% accuracy in validation, and 100% accuracy in training.

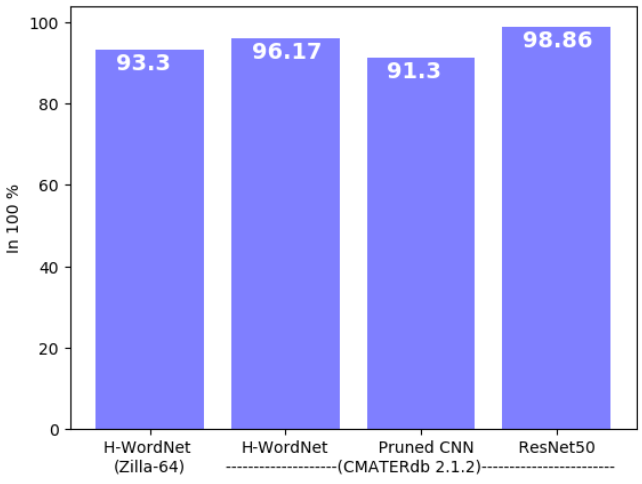


Fig. 7: Comparison of CNN models accuracy between 'Zilla-64' and 'CMATERdb 2.1.2' datasets.

The holistic-based H-WordNet [25], Pruned CNN model [26], and ResNet50 [27] clearly show good performance on CMATERdb 2.1.2. Alongside, H-WordNet also shows comparatively excellent performance on the Zilla-64 dataset.

The Average-Pooling layer in the last convolutional block of the H-WordNet model helps to achieve a promising performance instead of using a Max-Pooling layer. The accuracy can be higher with the increasing number of samples per class in the proposed dataset. The testing accuracy (93.30%), validation accuracy (96.21%), and training accuracy (100%) are not similar because of selecting the images using a random process for the experiment.

V. Conclusion

This research introduces a new challenging dataset in the Bangla handwritten recognition area. A recognition model has been applied to this dataset to prove the stability and robustness of the proposed dataset. CNN shows an excellent success rate in handwritten word recognition system as a holistic approach. By this work, a variation of writing styles has been added to Bangla literature. Primarily, the dataset is most useful for new researchers in the BHWR area and for testing a deep learning model efficiency. As the dataset is well-labeled, it can be helpful for the biometric recognition of a person. Furthermore, it can be applied to Bangla's handwritten city name recognition in postal automation. The dataset will surely help the researchers, increasing the contributions for the computer vision and pattern recognition area. In prospects, we intend to extend our dataset, and also researchers can help us add more images to this dataset.

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